Element A

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I Problem statement

Blind and legally blind pedestrians in the United States risk colliding with head- and chest-level obstacles during everyday walking, both indoors and outdoors, when hazards extend above a cane's reach. In a mostly U.S. survey, 13% reported head-level collisions at least monthly. These collisions can cause injuries and limit independent travel.[1]

II Justification

A cane may pass beneath a hanging branch, leaving the person's upper body exposed. Projecting signs, open cabinets, and shelf edges create similar hazards. A missed obstacle can cause injury and reduce a person's willingness to travel independently.

Manduchi and Kurniawan's 2009 survey included 307 adults, 289 of whom lived in the United States. Of the full sample, 300 answered the head-collision question. Medical consequences occurred in 23% of reported head-level accidents. Within that subset, 60% required professional medical assistance. Walking habits changed after 43% of accidents, and confidence in independent travel was affected in 26%. These results describe a voluntary sample, not the current rate for all blind pedestrians.[2]

RNIB's 2025 report on pedestrian travel in the United Kingdom describes injuries involving overhanging greenery and temporary obstructions. It also links these experiences to changes in confidence. The report provides recent context, but it does not establish U.S. collision rates. (RNIB, 2025)[3]

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[1] Manduchi and Kurniawan (2010), pp. 2, 4–6. The study measured head-level accidents. Chest-level hazards are an additional focus of this project.

[2] Manduchi and Kurniawan (2010), pp. 2, 4–6. The sample leaned toward younger, experienced travelers and included a relatively high share of guide-dog users.

[3] RNIB (2025), In my way, section 8, Injury.

The wearer's needs should guide the design of an additional warning device. Alerts must arrive early enough to allow a response and remain clear while walking. Comfort also matters because an irritating device may go unused. Orientation and mobility specialists can evaluate how the device fits existing travel skills. Family members and funders may assist with a purchase, but the wearer should judge its usefulness.

Existing products and patents

Existing products address some obstacles above ground level and provide useful comparisons. A new device therefore needs a clear advantage. This project investigates whether glasses can provide timely warnings while remaining comfortable and compatible with the wearer's usual mobility aid.

White canes and guide dogs are important starting points for that comparison. Manduchi and Kurniawan found no significant difference in head-accident frequency between cane and dog-guide users in their sample. This does not mean those aids are ineffective. It means an additional device should be evaluated for the protection it adds while the person continues using their usual aid. (Manduchi & Kurniawan, 2010)

The WeWALK Smart Cane 2 detects obstacles through a sensor in its handle and can alert the user through vibration or sound. Its manual explains that the sensor's direction depends on how the cane is held and that some objects and environmental conditions affect detection. Glasses could make the sensing direction independent of cane position. However, they would move with the wearer's head, so looking sideways could create a different blind spot. (WeWALK, n.d.)

NOA by biped.ai is a shoulder-worn system using depth and infrared cameras. Its 2024.7 manual lists a range of 0.3 to 10 meters, with 170 degrees of horizontal coverage and 90 degrees

of vertical coverage. As a direct wearable competitor, NOA sets a useful comparison. The proposed glasses would need to offer a benefit that matters to users, such as improved comfort for a particular activity. (biped.ai, 2024)

Some camera glasses read text or describe scenes. RNIB distinguishes these information tools from mobility aids. Object identification and collision warnings serve different purposes: a warning device must detect a hazard in time for the wearer to respond, even if it cannot name the object. (RNIB, n.d.)

Bai and colleagues combined depth sensing and ultrasound in research glasses in 2017. Their 2019 system added motion sensing and object-category information. Testing involved 20 visually impaired participants in an office setting. These studies support feasibility under some conditions, but they do not establish outdoor performance or readiness for independent travel for the proposed design. (Bai et al., 2017, 2019)[4]

The patent search found closely related ideas. WO1997001320A1, published in 1997, describes ultrasonic spectacles that communicate direction and distance through vibration. CN105342817A, published in 2016, describes glasses with upward- and downward-oriented ultrasonic sensors and an alert unit. Obstacle-detecting glasses are therefore an existing idea. Element B should compare specific design choices and examine the patent claims in greater detail. (OBRIRA GmbH, 1997; Shanghai Kanglong Baokang, 2016)[5]

[4] Bai et al. (2017, 2019), arXiv versions. Product specifications above come from manufacturer documentation and should not be treated as independent test results.

[5] WO1997001320A1 and CN105342817A, abstracts and descriptions. A published patent documents an invention; it does not prove its performance or establish clearance for a new product.

III Knowledge of the market

Users buyers and market size

The initial target users are U.S. adults who are blind or legally blind, use a cane or guide dog, and want additional warning about upper-body obstacles. Relevant trips include walking to school, work, stores, or transit stops. Glasses may suit some users, while others may prefer a chest clip, a cane attachment, or no additional device.

CDC's modeled estimates for 2024 put the U.S. population with best-corrected visual acuity loss at 7.15 million, including about 0.71 million people with acuity-defined blindness. These figures show that vision loss affects a sizable population, but they are not a count of potential buyers. The estimates do not include visual-field or contrast-sensitivity loss, and their categories do not cover everyone who is legally blind. (CDC, 2026)[6]

The likely customer group is much smaller than the national population with vision loss. Potential users need a relevant travel need, a comfortable fit, and a benefit that justifies the cost. The national population figure therefore cannot serve as a sales forecast. A more useful estimate would identify people reachable through local organizations and determine how many want to try the concept.

The user and buyer may be the same person. In other cases, a family member or organization may help pay. Employers and benefit programs are possible funding routes, but eligibility and purchasing agreements have not been established. The wearer's preferences should guide the design because the wearer must manage its weight, alerts, and maintenance.

The market is meaningful because upper-body obstacles can affect safety and everyday independence. Organizations serving people with vision loss offer a possible way to reach

[6] CDC (2026), VEHSS modeled estimates, key findings and vision loss categories. The model defines blindness using acuity of 20/200 or worse in the better-seeing eye.

potential users. Second Sense in the Chicago area, for example, provides individualized assistive-technology training. It is a potential contact for advice and permission to share a research invitation, although no partnership has been established. (Second Sense, n.d.)[7]

Competition and the proposed place in the market

WeWALK and NOA are the main direct comparisons. Relevant measures of competition include obstacle-detection reliability, warning time, unnecessary alerts, and comfort during use.

Reliable detection is a major barrier to entering this market. A demonstration using a large wall does not establish performance around thin branches, angled surfaces, or different materials and lighting conditions. The device must also balance frame weight, battery life, and accessible controls. Compatibility with prescription eyewear and hearing devices adds further constraints. Clear test results and an explanation of detection limits are essential for earning users' trust. Existing patents, repairs, and support costs also affect development.

Survey findings and further research

The survey summary contains 38 responses: 23 from people with low vision, 3 coded as blind under "Other," and 12 from family members, friends, or caregivers. Questions address travel experiences, current mobility tools, and preferences for an additional warning device. Figures 1–4 summarize the answers most relevant to the proposed concept.

These results describe a small sample rather than U.S. blind and low-vision pedestrians as a whole. The totals combine firsthand answers with supporter responses, so differences between the groups cannot be assessed. The survey also does not establish willingness to pay or purchase intent.

[7] Second Sense (n.d.), Technology solutions and training. This report does not claim a partnership or completed recruitment.

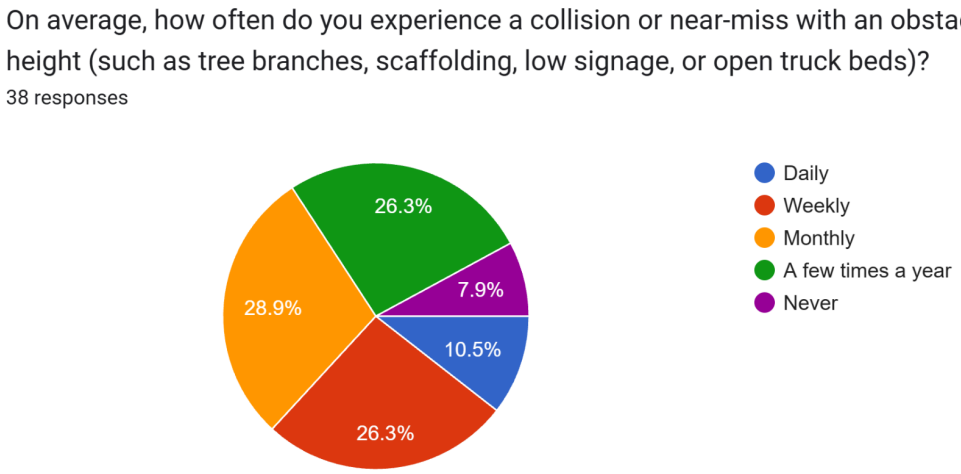


Figure 1. Reported frequency of collisions or near misses with obstacles above waist height. Source: Team survey summary, 38 responses.

Figure 1 shows that 25 of 38 respondents (65.8%) selected daily, weekly, or monthly encounters. Because the question combines collisions and near misses, this percentage is not an injury or collision rate. It cannot be compared directly with the published head-collision statistic in the problem statement.

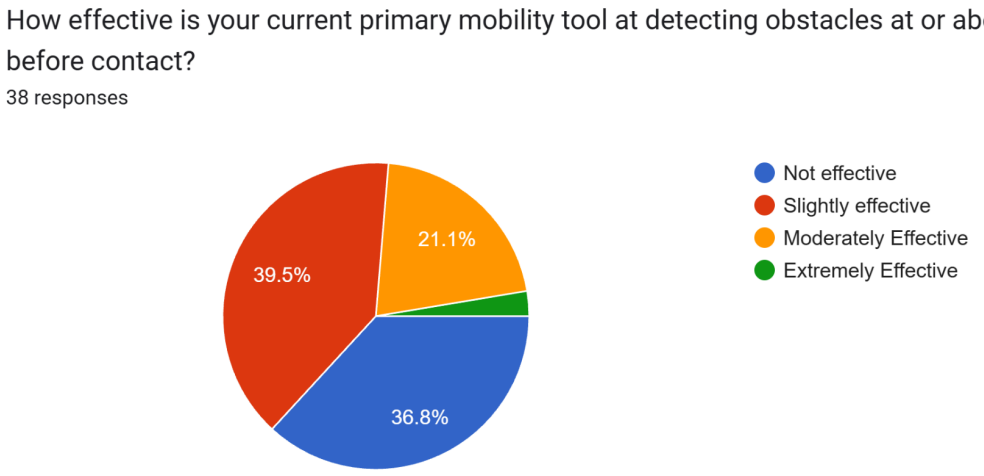


Figure 2. Ratings of current mobility tools for detecting obstacles at or above chest level before contact.

Source: Team survey summary, 38 responses.

In Figure 2, 29 of 38 respondents (76.3%) rated their current tool as not effective or slightly effective for detecting chest-level obstacles. This suggests an area where an additional warning device could help. These are reported ratings, however, rather than measurements of a tool's detection accuracy.

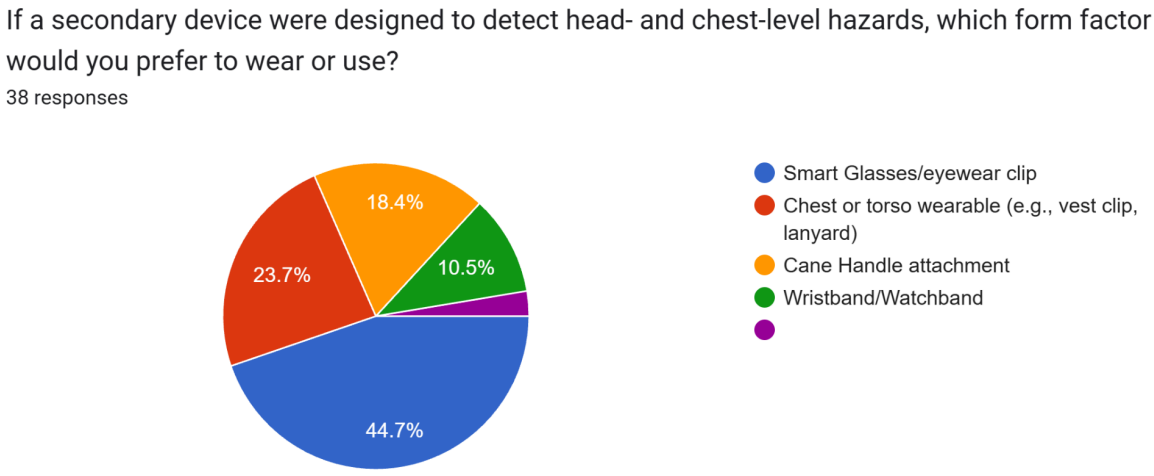


Figure 3. Preferred form for a secondary device that detects head- and chest-level hazards. Source: Team survey summary, 38 responses.

Figure 3 shows that glasses or an eyewear clip received the most selections: 17 of 38 (44.7%). This supports exploring the glasses concept, although it is not a majority. The other 21 respondents selected another option. Fit and comfort should therefore be compared across device forms before selecting a design.

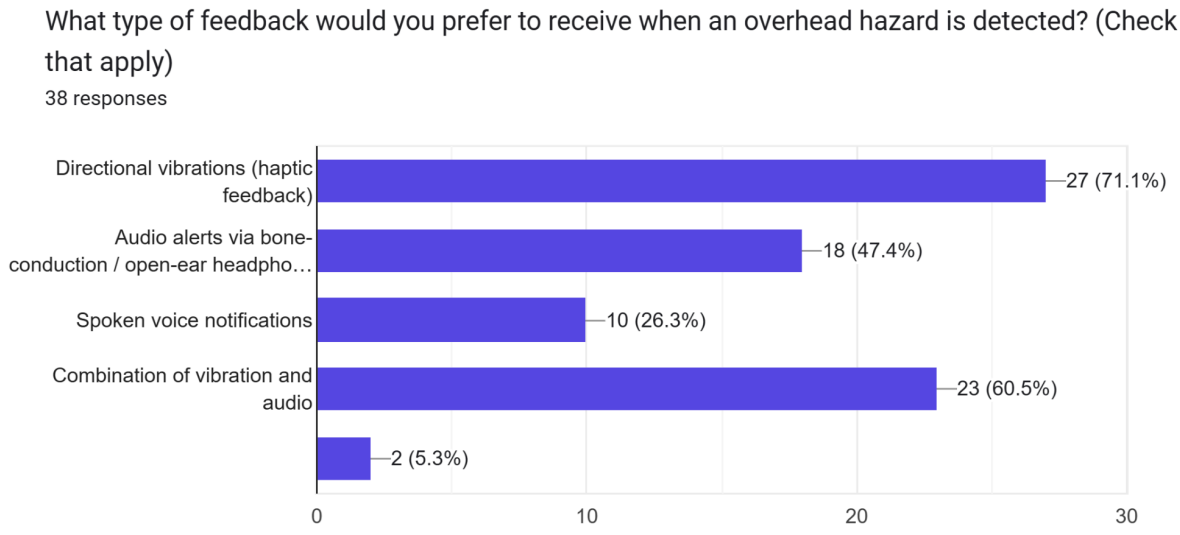


Figure 4. Preferred warning feedback. Multiple selections were allowed; percentages do not total 100%.

Source: Team survey summary, 38 responses.

As Figure 4 shows, directional vibration was selected by 27 respondents (71.1%), and a combination of vibration and audio by 23 (60.5%). Both options merit testing. Multiple selections were allowed, so the groups overlap and cannot be added together. These preferences do not establish which alert is easiest to notice while walking.

Follow-up research should separate firsthand answers from supporter observations and ask about collisions and near misses separately. The confidence scale also needs clear endpoint labels. A review by 5–8 potential users and 1–2 orientation and mobility professionals could improve the questions before broader distribution. The course guidance suggests ideally reaching 200–300 relevant respondents.

Later testing should measure the share of obstacles detected, warning time before predicted contact, and unnecessary alerts per minute. Results should be separated by obstacle height, material, size, and head direction. Practical measures include frame weight, wear time, fit

complaints, and user ratings after a defined task. Testing should also check interference with cane or guide-dog use and the response to low battery or a covered or unavailable sensor.

IV Conclusion

A missed upper-body obstacle can cause injury and change travel habits. Published research supports the need for better warnings, while existing products and patents provide a starting point for development. The next stage should determine whether glasses can deliver timely alerts comfortably and without distraction. Feedback from intended wearers and controlled testing should guide the choice of device form.

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Appendix A Survey summary

The team supplied the following aggregate counts for 38 survey responses. Figures 1–4 present the encounter, effectiveness, device-form, and feedback results. Percentages use 38 as the denominator and are rounded to one decimal place.

Respondent experience: low vision, 23 (60.5%); family member, friend, or caregiver, 12 (31.6%); blind, coded under "Other," 3 (7.9%).

Confidence in unfamiliar outdoor environments, on the form's 1–5 scale: 1, 5 (13.2%); 2, 11 (28.9%); 3, 13 (34.2%); 4, 7 (18.4%); 5, 2 (5.3%). The mean is 2.74. Without clear scale endpoints, this average should be interpreted cautiously.

Mobility tools used: cane, 27 (71.1%); human guide, 12 (31.6%); other, 11 (28.9%); guide dog, 7 (18.4%). Multiple tools could be selected, so percentages exceed 100%.

Figure 1 counts: Daily, 4 (10.5%); Weekly, 10 (26.3%); Monthly, 11 (28.9%); A few times a year, 10 (26.3%); Never, 3 (7.9%).

Figure 2 counts: Not effective, 14 (36.8%); Slightly effective, 15 (39.5%); Moderately effective, 8 (21.1%); Extremely effective, 1 (2.6%).

Figure 3 counts: Glasses or eyewear clip, 17 (44.7%); Chest or torso wearable, 9 (23.7%); Cane handle attachment, 7 (18.4%); Wristband or watchband, 4 (10.5%); No response, 1 (2.6%)

Figure 4 counts: Directional vibration, 28 (73.7%); Open-ear audio alerts, 18 (47.4%); Spoken notifications, 10 (26.3%); Vibration and audio, 22 (57.9%); Other, 2 (5.3%).